POLITECNICO DI MILANO

Scuola di Ingegneria Industriale e dell'Informazione

Master Degree in Mechanical Engineering
Course of Advanced Dynamics of mechanical systems

REPORT ASSIGNMENT 1B

Professor: Roberto Corradi

Tutor: Ing. Ivano La Paglia

Students:

Concari Mattia

D'Avino Gianfranco

Miuccio Michela Paola Benedetta

Rizzo Chiara

A.A. 2023-2024

Sommario

INTRODUCTION 3

EXPERIMENTAL SETUP 3

EXPERIMENTAL FREQUENCY RESPONSE FUNCTION 3

NUMERICAL FRFS: MODEL PARAMETERS IDENTIFICATION METHOD 4

FIRST VIBRATION MODE 6
SECOND VIBRATION MODE 7
THIRD VIBRATION MODE 8
FOURTH VIBRATION MODE 10

MODE SHAPE IDENTIFICATION 11

FIRST MODE SHAPE 11
SECOND MODE SHAPE 12
FOURTH MODE SHAPE 12

Introduction

The aim of this project is to analyze railway noise, a type of environmental noise caused by the operation of rail vehicles. The target of this section of the assignment is to perform experimental modal analysis of a light-rail wheel in order to identify its modal parameters. In particular, the discussion is focused on the axial vibrations because they lead to the perturbation of the air and, consequently, to sound radiation.

Experimental setup

The resilient wheel of a light-rail vehicle is composed by three parts: the wheel rim, the wheel web and the wheel hub which connects the wheel and the shaft; in addition, a rub element is interposed between the first two parts in order to damp the system. The wheel is suspended through 4 elastic supports in order to reproduce the boundary conditions obtaining, so, a free-free system. It has been decided to put four springs with the same nominal stiffness in a parallel arrangement in order to bear the weight and also to respect the symmetry of the system. These elements modify the system adding vibration modes: at low frequencies, the system is a rigid body that oscillate on springs along the vertical direction strongly decoupling rigid and flexible modes: to obtain an optimal result, it is required to choose springs such that at high frequency the rigid ones can't be seen in the spectrum of the FRFs.
The input is given through a dynamometric impact hammer that apply an axial load reproducing an impulsive excitation that theoretically excites all the frequencies in the frequency range considered at the same time: the harder the tip of the hammer is, the wider the range of excited frequencies will be. A load cell is arranged inside the hammer to measure the force applied on the system, whereas the generated axial vibrations of the wheel rim which can be sensed by 12 piezoelectric accelerometers located on the right half of the wheel, with a regular angular spacing of 15°, as shown in the picture below:

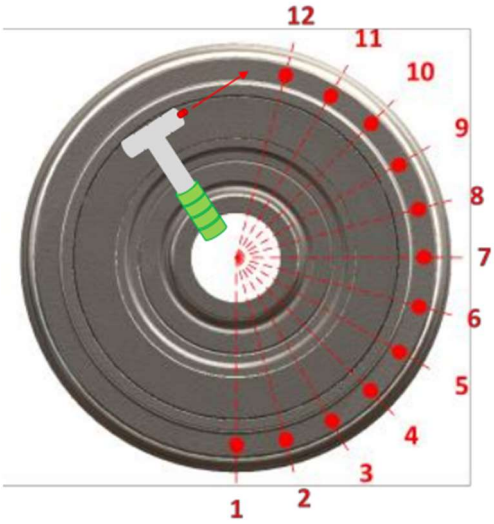

Figure 1 : 12 measurement positions on the wheel

Experimental Frequency Response Function

In the case taken into consideration the output is expressed as ***inertance,*** so the FRFs represent the ratio between acceleration and force. Several repetitions of the experiment are conducted in order to reduce the noise and obtain a mean value and the resulting load and the acceleration signals are measured by the load cell and the accelerometers in the time domain. Then, thanks to the Fourier Transfer function the signals are moved in frequency domain evaluating the auto and cross spectra.
Twelve distinct FRFs are obtained from the 12 measurement positions and the data used to draw the FRFs have been collected in the file *"Data.mat"*, that provides:

- **freq**: vector of frequencies of interest with a resolution of 0.333 Hz;
- **frf :** complex inertance frequency response function of all 12 output;
- **cohe:** coherence function, which estimates how good the measurements are.

The FRFs obtained are shown in the following figure:

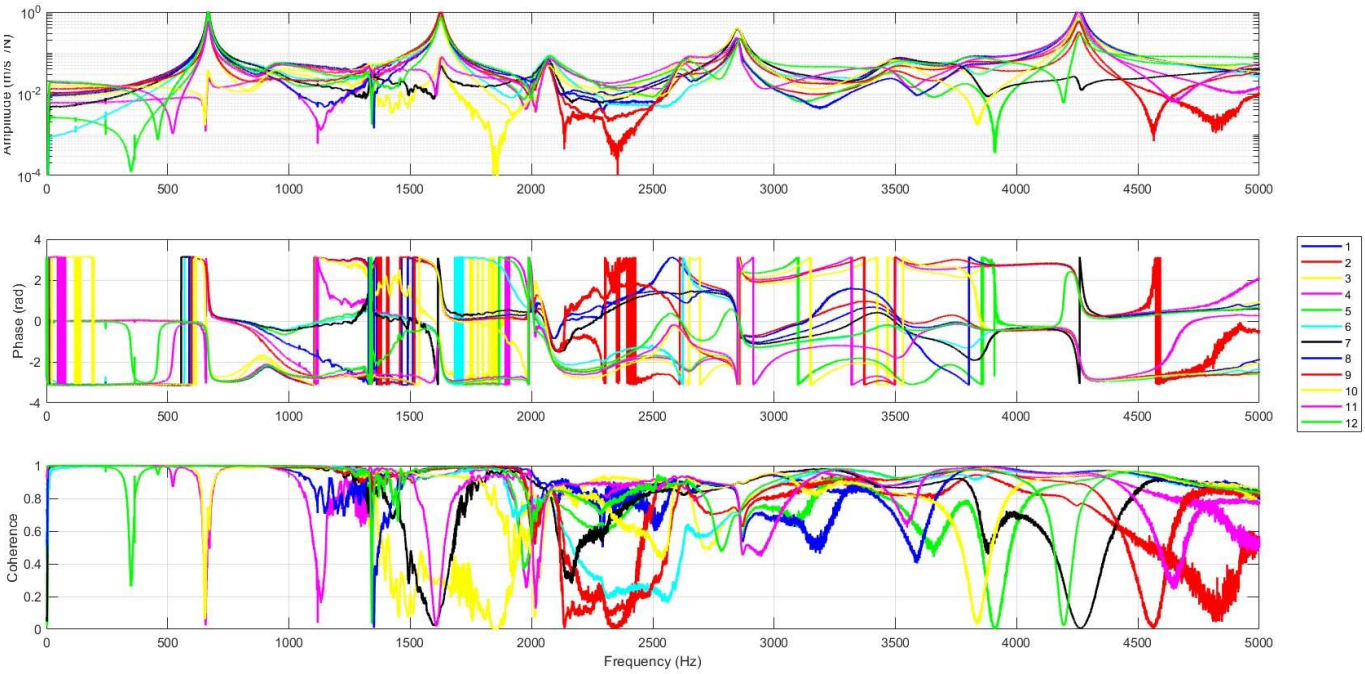

Figure 2: 12 experimental FRFs

The curves collected in the first box represent the amplitudes of the FRFs: the four higher peaks are associated with axial vibration modes, whereas the other smaller peaks are probably associated with radial modes. On the other hand, the curves represented in the third box represent the coherence, which shows how good the data are in a specific frequency domain: the results are very good up to 2000Hz approximately since the curve remains in the around of 1 showing some drops in correspondence of the antiresonances, which are not problematic because at that particular frequencies the sensor is not vibrating, so it doesn't measure anything.

Numerical FRFs: Model parameters identification method

Starting from the experimental FRFs it is possible to compute the numerical form of the FRFs, which are represented by the function:

$$G_{jk}^{NUM}(\Omega) = \sum_{i=1}^{n} \frac{A_{jk}^{(i)}}{-\Omega^2 + 2j\xi_i\omega_i\Omega + \omega_i^2} + \frac{R_{jk}^{L}}{\Omega^2} + R_{jk}^{H}$$

In the function above:

- The sum represents the contribution of all the (i) modes in the frequency range of interest $[\Omega_{min};\ \Omega_{max}]$;
- The residuals are introduced to consider also the influence of the modes not included in the selected frequency range, in particular R_{jk}^{L} stands for low frequency modes (that are expected to be in the seismic region in the considered range) whereas R_{jk}^{H} stands for high frequency modes (that are expected to be in the quasi-static region in the considered range);
- The unknowns are: ξ_i, ω_i, $A_{jk}^{(i)}$, R_{jk}^{L} and R_{jk}^{H}.

In the case taken into account it has been chosen n=1, meaning that in the frequency range of interest only one mode shape (the one resonating) can be considered to obtain an acceptable approximation of the FRF. For each vibration mode (corresponding to a single resonance peak), considering all 12 experimental FRFs, the damping ratio ξ_i and the natural frequency ω_i do not change since they do not depend on the specific combination of input and output locations, whereas the other terms vary with each FRF. For this reason, if N FRFs are taken into consideration, the number of unknowns for a single mode is equal to (2+5N) and they can be collected in a vector of unknown parameters.

At this point it is necessary to estimate the modal parameters that best fit the experimental data relying on a least square minimization procedure, using the *lsqnonlin* function in Matlab. Considering:

- M as the the length of the frequency vector, with Ω_s varying in the range $[\Omega_{min};\ \Omega_{max}]$;
- N as the number of the experimental FRF;

The mathematical expression considered for a single FRF is the following:

$$G_r^{NUM}(\Omega_s) = \frac{A_r^{(i)}}{-\Omega_s^2 + 2j\xi_i\omega_i\Omega_s + \omega_i^2} + \frac{R_r^L}{\Omega_s^2} + R_r^H$$

the error function to be minimized corresponds to the squared difference between two complex numbers evaluated for all the Ω_s and all experimental FRFs:

$$\epsilon = \sum_{r=1}^{N}\sum_{s=1}^{M} Re\left(G_r^{EXP}(\Omega_s) - G_r^{NUM}(\Omega_s)\right)^2 + Im\left(G_r^{EXP}(\Omega_s) - G_r^{NUM}(\Omega_s)\right)^2.$$

It is possible to proceed by considering once mode per time and defining the initial guesses for the unknowns required by the *lsqnonlin* function:

- ω_0 has been guessed identifying graphically the resonance peak in the frequency range of interest for the considered vibration mode;
- ξ_0 has been predicted through the slope of the phase diagram and has also been checked with the half power points method:

$$\xi_0 = -\frac{1}{\omega_i * \left.\frac{\partial\phi}{\partial\Omega}\right|_{\Omega=\omega_i}};$$

- Each A_{jk0} has been guessed evaluating the imaginary part of the FRFs in correspondence of the resonance peak ($G_{jk}^{EXP}(\omega_0)$) and multiplying it for the factor:

$$A_i = -Im\left(FRF_i^{EX}(\omega_0)\right) * 2 * \omega_0^2 * \xi_0$$

- The residuals R_0^L and R_0^H have been imposed null - both in the real and imaginary parts - because they ideally do not have any influence.

The vector $\underline{x_0}$ collecting the unknowns is, then:

$$\underline{x_0} = \left[\omega_0, \xi_0, A_{jk}(:)', zeros(1,12), zeros(1,12), zeros(1,12), zeros(1,12)\right],$$

where A_{jk} is a vector which contains the 12 initial values.

Starting from these values, it is possible to solve a minimization problem through the Matlab function *lsqnonlin*, which gives back a vector with the parameters requested to compute the G_r^{NUM}.

As an example, the table below shows the comparison between the initial guessing values of the frequencies and damping ratios and the ones obtained by the numerical method:

	INITIAL VALUES	IDENTIFIED VALUES
1st mode shape	f_1 = 666.9911 Hz ξ_1 = 0.008	f_1 = 666.99 Hz ξ_1 = 0.8%
2nd mode shape	f_2 = 1626 Hz ξ_2 = 0.006	f_2 = 1625.98 Hz ξ_2 = 0.6%
3rd mode shape	f_3 = 2849 Hz ξ_3 = 0.0064	f_3 = 2828.96 Hz ξ_3 = 0.64%
4th mode shape	f_4 = 4255.9 Hz ξ_4 = 0.0036	f_4 = 4255.94 Hz ξ_4 = 0.36%

The coupled values are almost equal for each mode shape taken into consideration, but the quality of the identification could be better analyzed in the figures below, which show the results obtained for all the four vibration modes comparing the identified FRFs and the experimental ones.

First vibration mode

The first vibration mode presents a resonance peak around 650Hz, so the frequency range considered is [600:700] Hz.

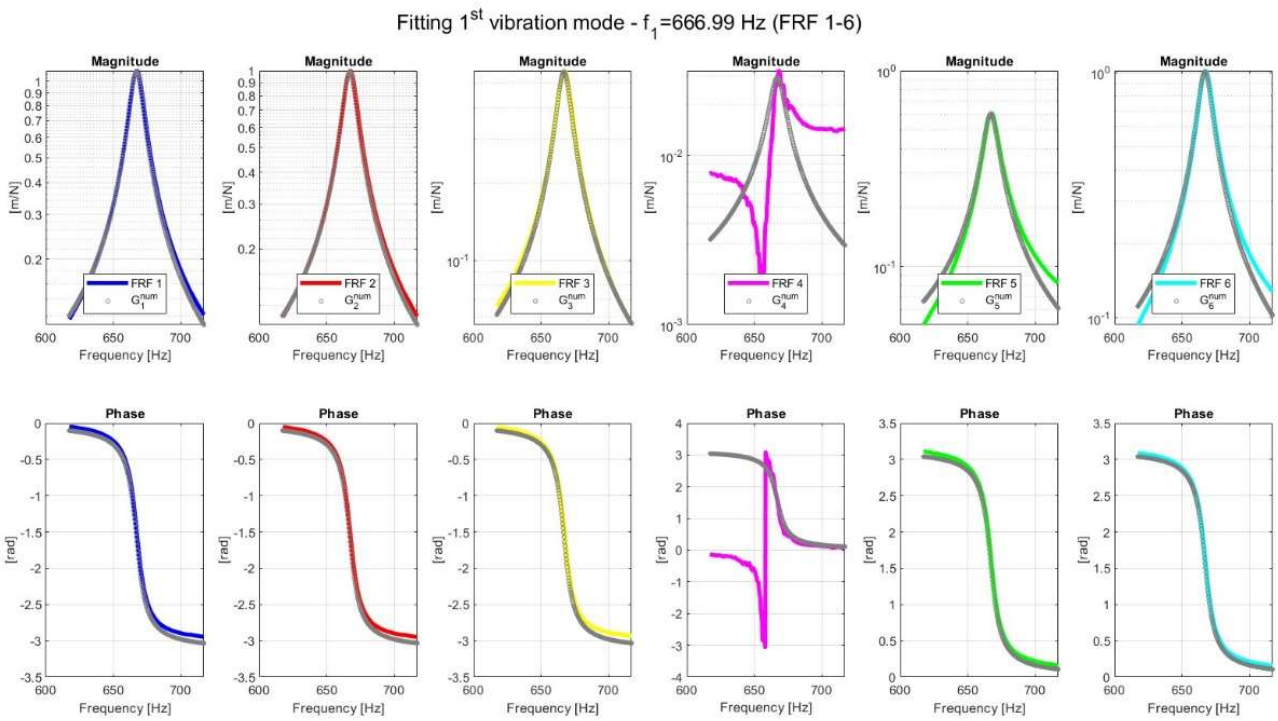

Figure 3: Fitting of the 1st vibration mode, FRFs from 1 to 6

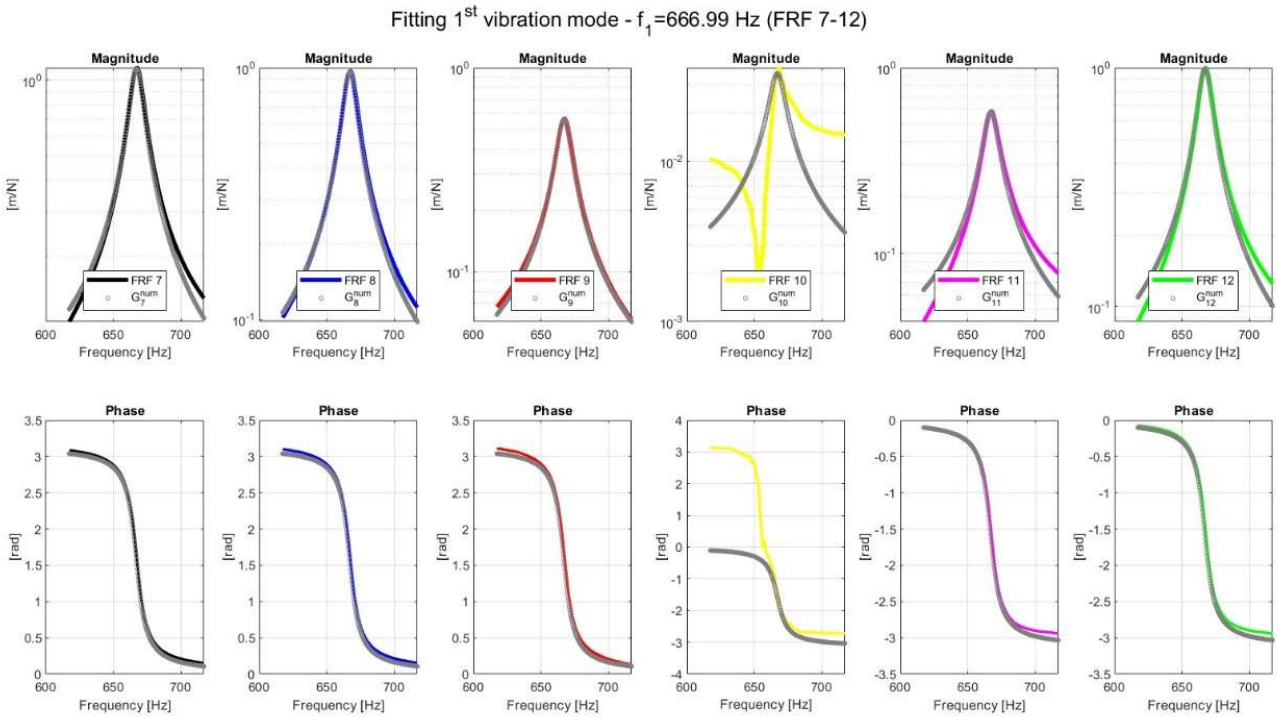

Figure 4: Fitting of the 1st vibration mode, FRFs from 7 to 12

The identified FRFs overlap the experimental ones in almost all cases, except for the FRFs number 10 and 4. The reason of this phenomenon is linked with the behavior of these two function. They don't present a resonance peak at 650 Hz, the frequency of interest. The corresponding sensors for those FRFs are close to a node of the first vibration mode. They present an antiresonance and immediately after a resonance this phenomenon happens when a sensor in positioned near a node of vibration.

Second vibration mode

The second vibration mode presents a resonance peak around 1626 Hz, so the frequency range considered is [1576:1676] Hz.

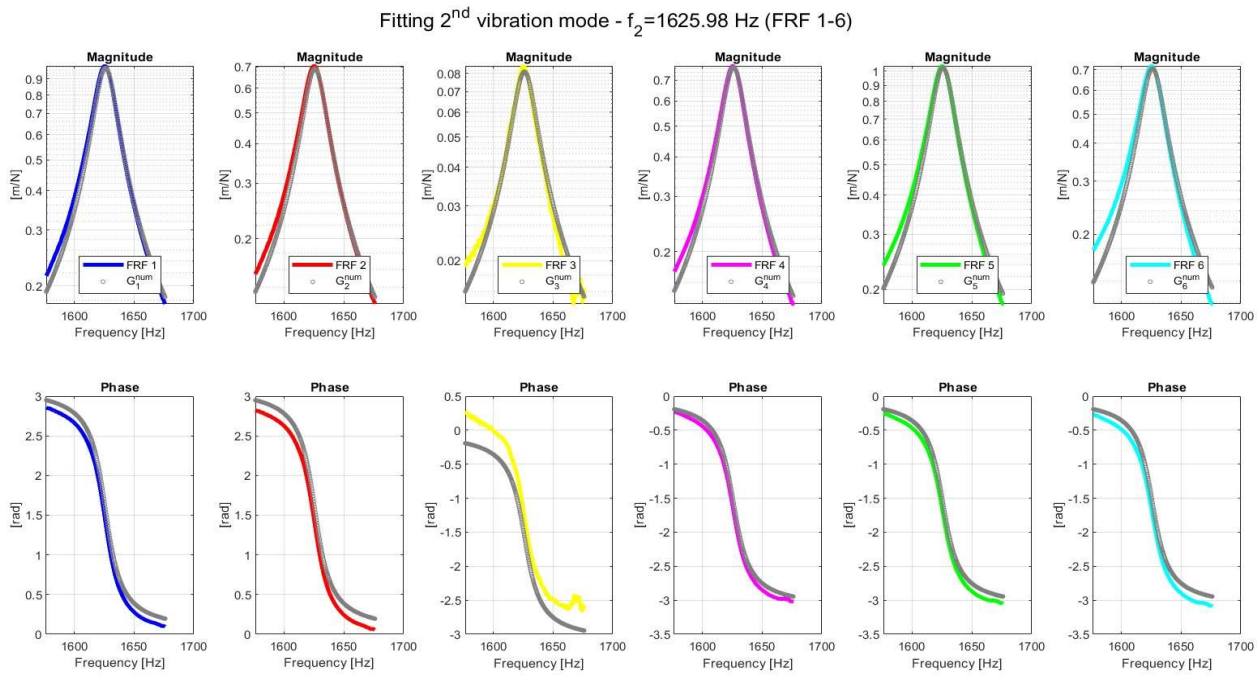

Figure 5: Fitting of the 2nd vibration mode, FRFs from 1 to 6

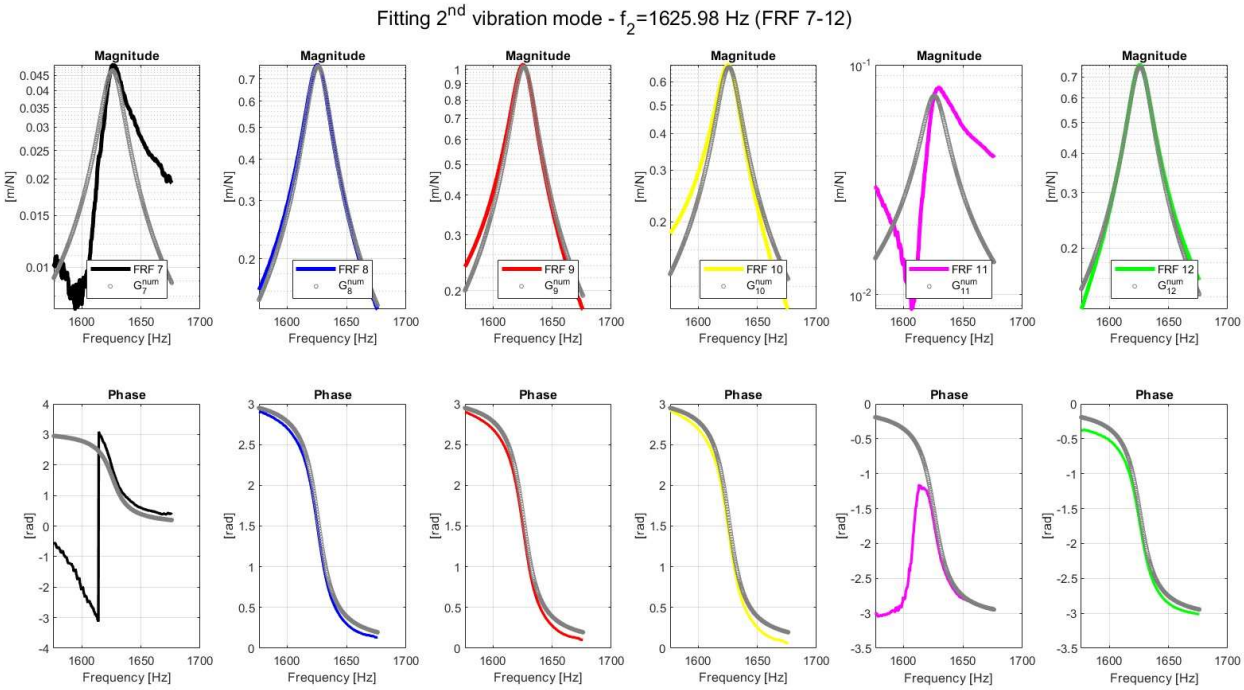

Figure 6: Fitting of the 2nd vibration mode, FRFs from 7 to 12

In this case, the 7th and 11th numerical FRFs do not fit well the experimental ones: the reason depends on the position of the sensors. In fact, the two sensors taken into account are located near two nodes of the second vibration mode, so they present antiresonances and immediately after resonances: in these cases, the numerical method it is not particularly powerful and it is not able to fit these FRFs.

Third vibration mode

The third vibration mode presents a resonance peak around 2849 Hz, so the frequency range considered is [2799:2899] Hz.

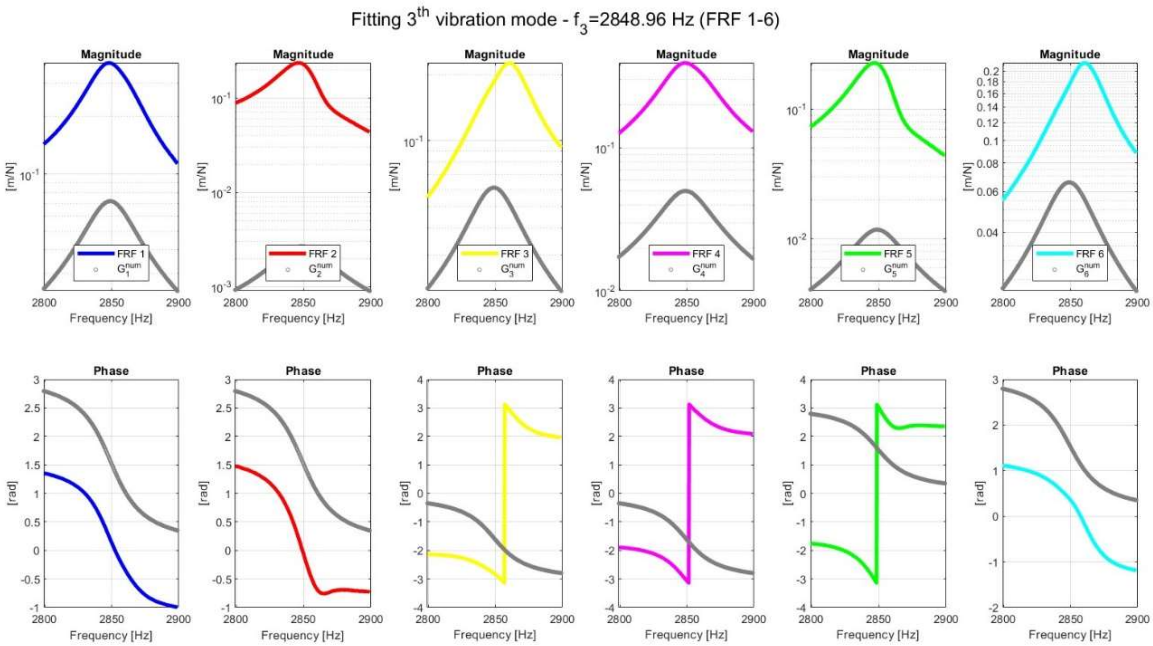

Figure 7: Fitting of the 3rd vibration mode, FRFs from 1 to 6

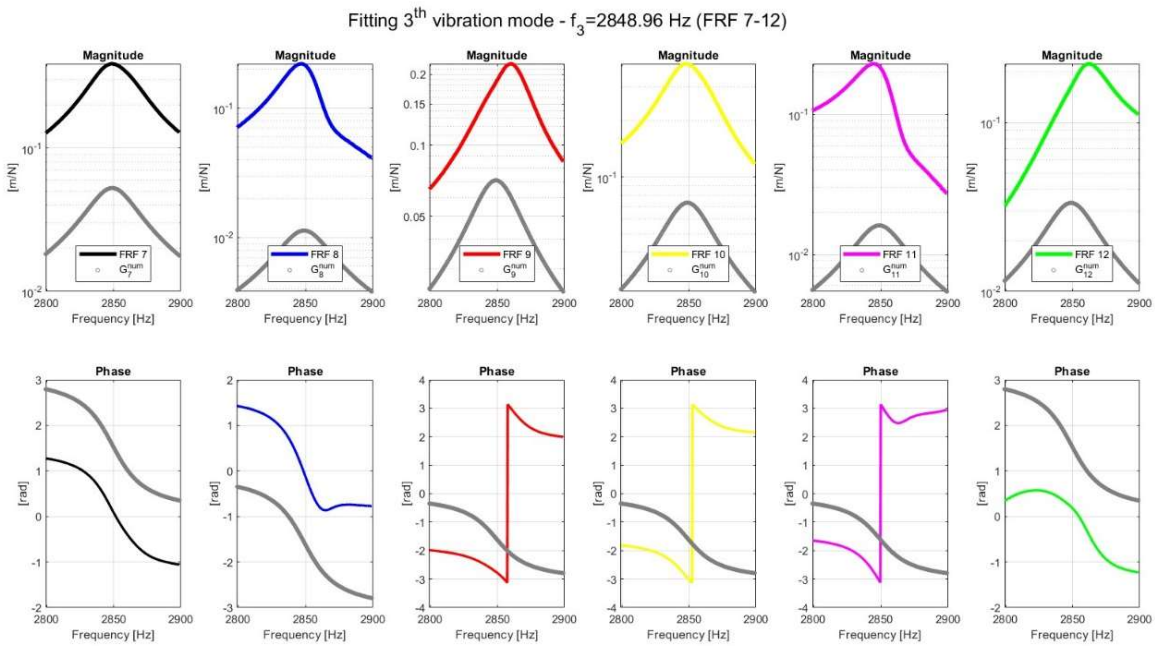

Figure 7: Fitting of the 3rd vibration mode, FRFs from 7 to 12

In this case, none of the experimental FRFs is adequately fitted by the numerical method: this phenomenon can be explained by the presence of three resonance peaks (Figure 8) in the range of frequency of interest, which make the used method not suitable in this situation. For a deeper comprehension, it is important to make some considerations about the used model:

$$G_{jk}^{NUM}(\Omega) = \sum_{i=1}^{n} \frac{A_{jk}^{(i)}}{-\Omega^2 + 2j\xi_i\omega_i\Omega + \omega_i^2} + \frac{R_{jk}^{L}}{\Omega^2} + R_{jk}^{H}$$

In the other vibration modes, the parameter n has been imposed to one because only one resonance peak is present in the considered frequency range, but in this case n should be imposed to 2 because two peaks are very close that they could be considered as unique resonance frequency. Therefore, $G_{jk}^{NUM}(\Omega)$ is obtained summing of two factors with different parameters ξ_i,ω_i, $A_{jk}^{(i)}$; also the additional initial conditions have to be inserted in the $\underline{x_0}$ vector.

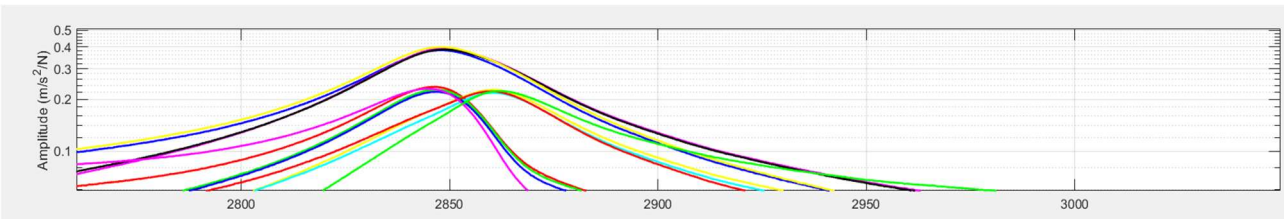

Figure 8: Three resonance peaks

The cause of the phenomenon is probably linked to the experimental setup: for this frequence the 12 sectors are excited in different ways so the structure is not symmetric. Probably, the four springs affect the wheel vibrating behaviour and two close vibration modes are excited simultaneously. It is a particular occurrence that could not happen if the input changes.

Fourth vibration mode

The fourth vibration mode presents a resonance peak around 4255 Hz, so the frequency range considered is [4205:4305] Hz.

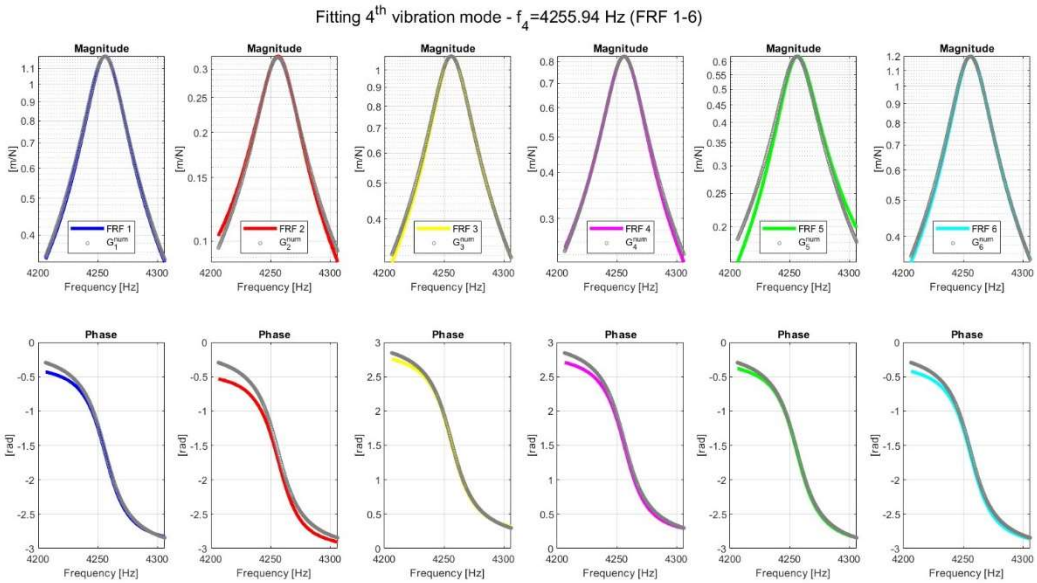

Figure 9: Fitting of the 4th vibration mode, FRFs from 1 to 6

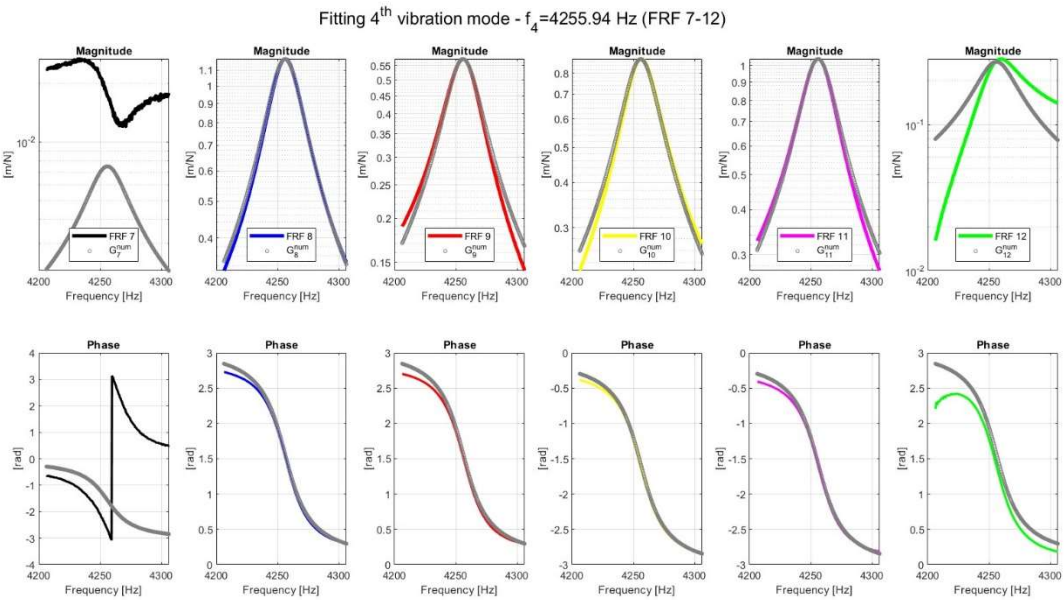

Figure 10: Fitting of the 4th vibration mode, FRFs from 7 to 12

Differenlty from the third vibration mode, the fourth one is perfectly fitted by the numerical method except for the 7th and 12th FRFs, which do not present a resonance peak at the frequence of interest. The reason is still linked to the experimental setup because the sensors 7 and 12 are close to two nodes of this vibration mode, so the responses present antiresonances and immediately after resonances.

Mode shape identification

Once the numerical FRFs have been obtained in correspondence of the resonance peaks thanks to the FRF-based multi-mode curve fitting method, it is possible to represent the corresponding mode shapes related to each axial vibration mode. Subsequently, it has been supposed that the first FRF is related to an accelerometer set on the bottom position of the wheel and the following 11 represent the other positions spaced by a 15° angle; the top of the wheel has been supposed to be the excitation point.
The expression of the mode shapes has been extrapolated from the imaginary part of the numerical G_{jk}^{NUM} obtained before:

$$\phi_i = Im(G_{jk}^{NUM}(\omega_i)$$

The value obtained for each FRF has been multiplied by a scaling factor (high enough to see the deformation) in the order of 10^3 . Then, the displacements calculated have been summed to the undeformed arbitrary radius in order to have a **qualitative** representation of the deformed tyre of the wheel for each vibration mode. Thanks to this procedure only the displacements in correspondence of the sensor are obtained, by using a Matlab interpolation function (*interp1*) a line which highlights the deformed shape is generated.
The other half of the wheel has been realized only by mirroring the measured part since the structure is considerable symmetric.

First mode shape

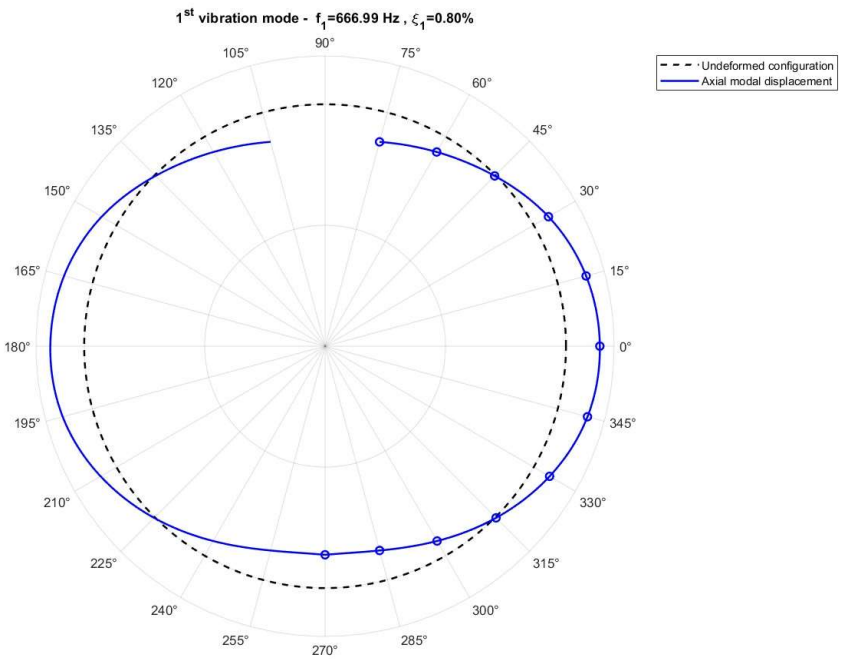

Figure 11: Deformed configurations for 1st vibration mode

In the graph it is possible to observe the first mode shape of the railway wheel, characterized by the natural frequency f_1 which has been identifed approximately equal to 667Hz: there are two nodal diameters that may lead to assess that the represented one is the second vibration mode. Probably, with the considered experimental setup, it is not possible to excite the first vibration mode. In fact, the following mode shapes (2 and 4) represent respectively the third and the fifth axial vibration modes, while the third mode shape is not showed because the numerical method used is not able to fit the FRFs, so the polar plot obtained does not reveal the real behaviour of the wheel.

Second mode shape

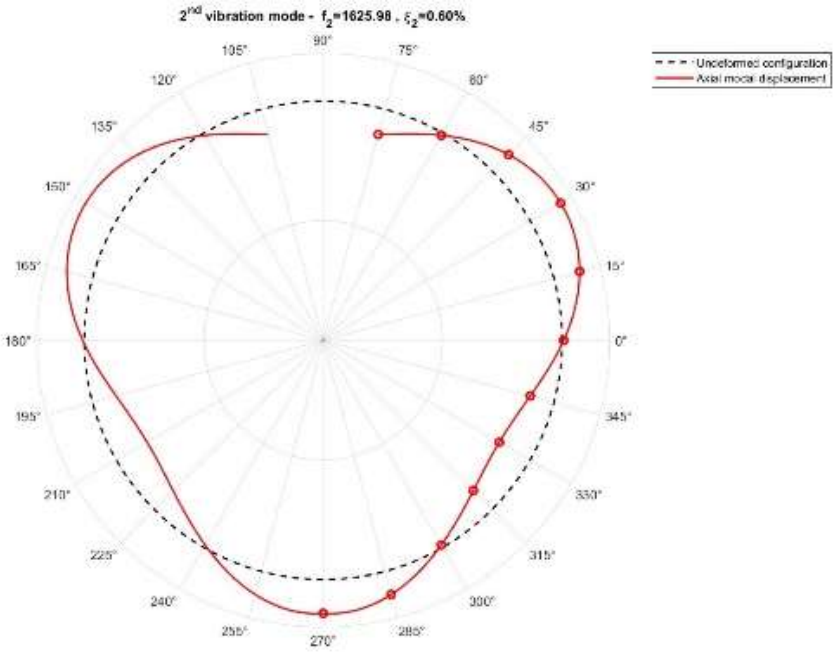

Figure 12: Deformed configurations for 2nd vibration mode

The second mode shape has an identified natural frequency of 1626Hz and a light damping factor that turns about 0.6%. As in the previous case, it is possible to assess that this is the third vibration mode due to the presence of three nodal diameters.

Fourth mode shape

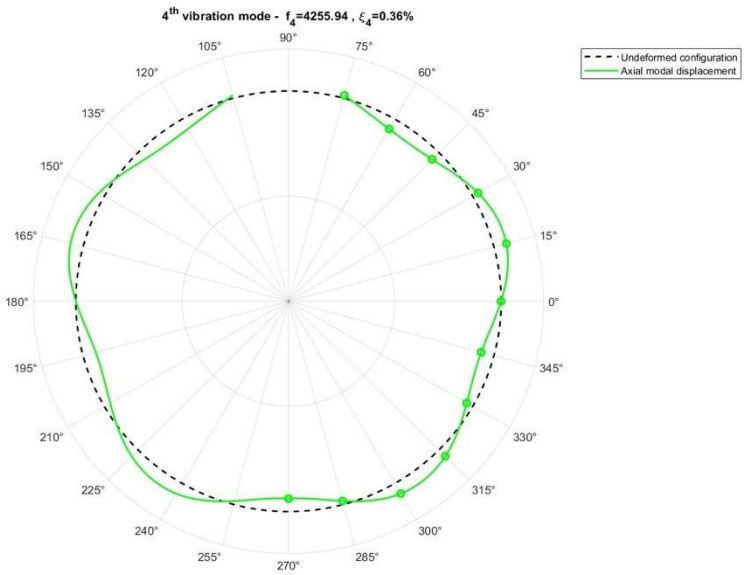

Figure 13: Deformed configurations for 4th vibration mode

The last axial vibration mode visible in the range of frequency about 5000Hz is the fifth (since there are five nodal diameters), which is featured by a natural frequency of 4255.94Hz and a decreasing damping factor with respect to the previous ones.